

UChicago REU Notes Week 4

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This note is mainly based on B. Guillou and J.P. May's *Equivariant iterated loop space theory and permutative G -categories*, trying to understand some machineries intuitively (may omit some details). The goal is understanding the categorical aspect and its relation to the more topological aspect.

Remark 0.1. Assume everything is compactly generated, so we have a natural homeomorphism $\mathcal{U}(X \times Y, Z) \cong \mathcal{U}(X, \mathcal{T}(Y, Z))$, and $\mathcal{T}(X \wedge Y, Z) \cong \mathcal{T}(X, \mathcal{T}(Y, Z))$. Here, $\mathcal{U}$ are compactly generated spaces, $\mathcal{T}$ are based spaces.

For basic reviews of G -spaces, I'll type up the notes later.

Here, we also assume V is a finite dimensional $\mathbb{R}$ -inner product space with a G -representation.

1 Basics of Equivariant Homotopy

This part cf. J.P. May's *Equivariant Homotopy and Cohomology Theory*.

2 Notations

Goal: Understand to construct genuine G -spectra. A naive G -spectra is a spectra with a G -action (supposedly the structure maps are all G -equivariant + homotopy equivalence, or homeomorphism). Seems like we require extra structures on the spectra. One of the distinction seems to be: naive G -spectra represents $\mathbb{Z}$ -graded cohomology theories, genuine G -spectra are used to represents cohomology theories for real representations (so it's also encoded using representations). I particular, understanding what is the equivariant infinite loop space machine (construct from G -space / category, forming a G -spectra).

Definition 2.1. Given a category $\mathcal{V}$, $X \in \mathcal{V}$ is a G -object if it has a group homomorphism $G \rightarrow \text{Aut}(X)$.

$\mathcal{V}_G$ denotes the category of G -objects, with morphisms being in $\mathcal{V}$. Each $\mathcal{V}(X, Y)$ has a natural G -action by conjugation, where $g \in G$ acts as $f \mapsto g \circ f \circ g^{-1}$.

$G\mathcal{V} \subseteq \mathcal{V}_G$ denotes the subcategory where homsets are limited to G -maps (so $G\mathcal{V}(X, Y) := \mathcal{V}(X, Y)^G$, fixed points of the homsets). Another notation is $G\mathcal{V} := (\mathcal{V}_G)^G$.

Definition 2.2. Given $X \in G\mathcal{T}$ (based G -space), assume inclusion of basepoint $* \rightarrow X$ is nondegenerated in the G -case, meaning it's a G -cofibration (satisfying G -homotopy extension property for G -equivariant maps, abbreviated as G -maps). In particular, all the homotopies in the extension property is required to be a G -map at all time t (another way of phrasing, interval I has trivial action, so action at slice $X \times t$ is equivalent to action on $X \times I$).

By chasing through the homotopy extension diagram (by restricting all homotopies to be G -homotopy), then inclusion of basepoint $* \rightarrow X^H$ becomes a cofibration with respect to G -maps (Note: Not necessarily a G -cofibration, because in general X^H is not closed under G -action). However, since all G -space is naturally an H -space (so is homotopy), then we can understand this as “ H -cofibration”, as H acts trivially on X^H :0

Definition 2.3. A G -map $f : X \rightarrow Y$ is a weak equivalence of G -spaces (or G -equivalence), if the fixed point maps $f^H : X^H \rightarrow Y^H$ are weak homotopy equivalences for all subgroups $H \leq G$.

Similar G -whiteheads theorem states if X, Y are G -CW complexes, then f is also a G -homotopy equivalence.

Definition 2.4. A small category $\mathcal{C}$ is a *topological category*, if the set of objects Ob and set of all morphisms Mor are topological spaces, such that the following four maps are continuous:

- Identity map $i : \text{Ob} \rightarrow \text{Mor}$ by $i(x) := \text{id}_x$ is a continuous map.
- Source map $s : \text{Mor} \rightarrow \text{Ob}$ by $s(f : X \rightarrow Y) := X$ is continuous.
- Target map $t : \text{Mor} \rightarrow \text{Ob}$ by $t(f : X \rightarrow Y) := Y$ is continuous.
- Composition $\circ : \text{Mor} \times_{t,s} \text{Mor} \rightarrow \text{Mor}$ is continuous.

Which, the last one is collecting all pair (X, Y) with morphism $f : X \rightarrow Y$ between them.

In particular, s, t are retracts of i , showing i is an embedding. (check <https://mathoverflow.net/questions/85717/segals-original-definition-of-a-topological-category>)

There is another notion enriched over $\mathcal{T}$.

Notation 2.5. Let Cat denotes the category of small topological categories, which Cat_G denotes the topological G -categories (meaning G acts on both Ob, Mor such that $i, s, t, \circ$ are all G -maps), an equivalent formulation is a group homomorphism $G \rightarrow \text{Aut}_{\text{Cat}}(\mathcal{C}, \mathcal{C})$, by checking the definition.

Part of the reason we endow this definition, is so that for subgroup $H \leq G$, the H -fixed category $\mathcal{C}^H$ will make sense, namely we need the action on Ob, Mor to be compatible.

Remark 2.6. Given the classifying space functor B , it realizes any topological category $\mathcal{C}$ into $B\mathcal{C}$.

Question: What are the comparison between the two spaces?

Which, for some reason it's product preserving, and it sends objects in Cat to $\mathcal{U}$, $G\text{Cat}$ to GU . In particular, it preserves the operad action (as preserving product also preserves the action).

3 E_V Operads, V -fold Loop G -Spaces

3.1 Little Disks / Steiner Operad

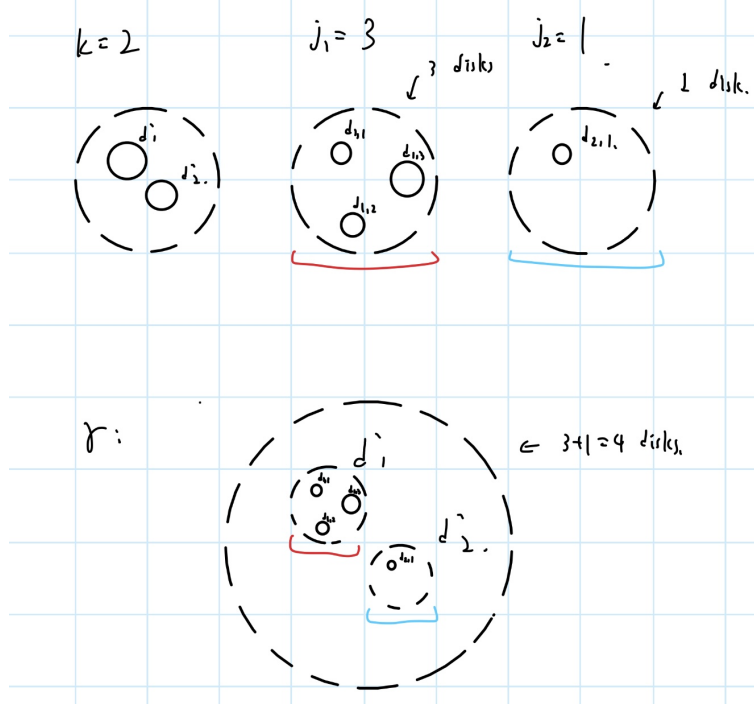
Definition 3.1. Given $D(V) \subset V$ the open unit disk (all $v \in V$ with $\|v\| < 1$), a *little V -disk* is an affine map $d : D(V) \rightarrow D(V)$, meaning all $d(v) = rv + v_0$, where $r \in [0, 1)$ and $v_0 \in D(V)$ represents the radius and the center.

Define group action $g \in G$ on collection $(gd)(v) := rv + gv_0$, which forms an evident G -action with restriction of action $G \rightarrow O(V)$ (the real unitary group of V).

Define the *little disk operad* by defining $\mathcal{D}_V(j) := \{(d_1, \dots, d_j) \mid d_i \text{ shares empty image intersection}\}$. Geometrically, it's all the possible arrangement of j non-overlapping disks in $D(V)$. Define the evident structure map $\gamma : \mathcal{D}_V(k) \times \mathcal{D}_V(j_1) \times \dots \times \mathcal{D}_V(j_k) \rightarrow \mathcal{D}_V(j)$ as follow:

$$\gamma((d'_1, \dots, d'_k); (d_{1,1}, \dots, d_{1,j_1}), \dots, (d_{j_k,1}, \dots, d_{j_k,j_k})) = ((d'_1 \circ d_{1,1}, \dots, d'_1 \circ d_{1,j_1}); \dots; (d'_k \circ d_{j_k,1}, \dots, d'_k \circ d_{j_k,j_k}))$$

Geometrically, it's as follow (for an example):



This is obvious a Σ -operad (as we defined it to be function composition). There is a correspondance of this definition to configuration space: Let $F(V, j) \subset V^j$ be the ordered configuration space of j -tuples of distinct points of V , with G acts on V^j diagonally (which restricts to an action on $F(V, j)$).

Theorem 3.2. *There is a $(G \times \Sigma_j)$ -homotopy equivalence $\mathcal{D}_V(j) \simeq F(V, j)$ for all $j \geq 0$.*

More explicitly, rescaling V to $D(V)$ (by a map $[0, \infty) \rightarrow [0, 1)$) so that $F(V, j) \cong F(D(V), j)$, then send $(d_1, \dots, d_j)$ to the centers $(v_1, \dots, v_j)$, and conversely given $(v_1, \dots, v_j)$, choose r to be half the smallest distance between these points, then send to the corresponding $(d_1, \dots, d_j)$ (each d_i with radius r and center d_i).

Which, since there is a free $(G \times \Sigma_j)$ -action (the space is not contractible or a local equivalence), so this is called an E_V -operad.

Definition 3.3. $\mathcal{C}_G$ an operad of G -spaces is an E_V -operad if there is a morphism of operads $\mathcal{C}_G \rightarrow \mathcal{D}_V$, such that each level $\mathcal{C}_G(j) \rightarrow \mathcal{D}_V(j)$ is a weak equivalence.

Which, given $V \subset W$ two G -orthogonal representations, in general there's no morphism of operad $\mathcal{D}_V \rightarrow \mathcal{D}_W$ that's compatible with suspension. (For details, the non-equivariant version has a suspension map between little n -cube operad to $(n+1)$ -cube operad, which is compatible with suspension; hence, taking the colimit results in an E_∞ -operad. In this case it doesn't work).

Definition 3.4. Let E_V be the space of embeddings $V \rightarrow V$ (not necessarily G -embeddings), together with G acting on E_V with conjugation.

Define $\text{Emb}_V(j) \subset E_V^j$ be the G -subspace with ordered j -tuples of embeddings $(\gamma_1, \dots, \gamma_j)$, such that each $i \neq j$ has disjoint image. The collection Emb_V naturally forms an operad in GU , by the existence of identity embedding, and similar notion of “composite embedding” as little V -disk operad.

Definition 3.5. Define $R_V \subset E_V = \text{Emb}_V(1)$ to be all distance-reducing embeddings, i.e. $f : V \hookrightarrow V$ such that $\|f(v) - f(w)\| \leq \|v - w\|$ for all $v, w \in V$. (Which, composition is closed).

Define a *Steiner path* $h : I \rightarrow R_V$ so that $h(1) = \text{id}_V$. Let P_V denotes the path space of Steiner paths, define $\pi : T_V \rightarrow R_V$ by $\pi(h) = h(0)$.

Define $\mathcal{K}_V(j)$ to be G -space of ordered j -tuples of Steiner paths $(h_1, \dots, h_j)$, such that each $\pi(h_i)$ has disjoint image, with unit in $\mathcal{K}_V(1)$ being the constant Steiner path. Define structure map $\gamma : \mathcal{K}_V(k) \times \mathcal{K}_V(j_1) \times \dots \times \mathcal{K}_V(j_k) \rightarrow \mathcal{K}_V(j)$ as follow: given $(h'_1, \dots, h'_k) \in \mathcal{K}_V(k)$, and $(h_{i1}, \dots, h_{ij_i})$, we define $h_{i1} \cdot h'_i : I \rightarrow R_V$ by sending t to $h_{i1}(t) \circ h'_i(t)$, which it's still a Steiner path. And, composition of such embedding preserves disjoint images, so this defines a map of operad by writing similar structure map.

There is a natural map of operad $\pi : \mathcal{K}_V \rightarrow \text{Emb}_V$ (by evaluating each Steiner path at 0). There is another map of “evaluation” at $\mathbf{0} \in V$, providing a $(G \times \Sigma_j)$ -map $c : \text{Emb}_V(j) \rightarrow F(V, j)$.

Theorem 3.6. *There is a weak equivalence of operad $\iota : \mathcal{D}_V \rightarrow \mathcal{K}_V$.*

(By showing $c \circ \pi : \mathcal{K}_V(j) \rightarrow \text{Emb}_V(j) \rightarrow F(V, j)$ is a homotopy equivalence, then use the lemma beforehand).

Intuitively, this can be thought of as: Embedding of disks is equivalent to embedding of V (by rescaling $D(V)$ back to the original space).

Part of the reason why $\mathcal{K}_V$ is invented, is because for inclusion $V \subset W$ of G -inner product spaces, the inclusion $\mathcal{K}_V \hookrightarrow \mathcal{K}_W$ (by $f : V \rightarrow V$ includes into $f \oplus \text{id} : W \rightarrow W$, where $W - V := V^\perp$ in W) of G -operads induce a map:

$$\Omega^V X \rightarrow \Omega^V (\Omega^{W-V} \Sigma^{W-V} X) \cong \Omega^W \Sigma^{W-V} X$$

And, this map preserves $\mathcal{K}_V$ -action, when given X a G -space.

3.2 Approximation Theorem

First, let $\mathbf{K}_V$ denotes the monad of the Steiner operad $\mathcal{K}_V$, which $\mathbf{K}_V(X) := \coprod \mathcal{K}_V(j) \times_{\Sigma_j} X^j(\sim)$ (quotient by commuting G -actions also?)

For the non-equivariant case, we have the following definition:

Definition 3.7. X is a *Hopf Space* if it has a continuous product $\mu : X^2 \rightarrow X$, together with a two-sided unit up to homotopy (meaning the “identity” $e \in X$ have $\mu(e, -), \mu(-, e)$ both homotopic to identity, or inducing a homotopy equivalence of X on itself). Which, $\pi_0(X)$ is a monoid.

X a Hopf Space is called *grouplike* if $\pi_0(X)$ is a group (meaning μ is associative up to homotopy, and all $\mu(x, -), \mu(-, x)$ are homotopy equivalence, which is only dependent on the path components).

Let X, Y be homotopy associative / commutative Hopf Spaces, with Y being grouplike, and $f : X \rightarrow Y$ be a Hopf Map (meaning the product map commutes with f up to homotopy), then f is a *group completion*, if $f_* : \pi_0(X) \rightarrow \pi_0(Y)$ is the group completion of commutative monoid $\pi_0(X)$, together with an isomorphism of homology algebra:

$$H_*(X)[\pi_0(X)^{-1}] \rightarrow H_*(Y)$$

In some sense, this can be thought of as Y completes X as a generalization of topological monoid.

Another definition of group completion relies on A_∞ -spaces, which is a space X over the Steiner operad $\mathcal{K}_{\mathbb{R}}$. An A_∞ -map is a map homotopic to a $\mathcal{K}_{\mathbb{R}}$ -space map, or the homotopy inverse of a map of $\mathcal{K}_{\mathbb{R}}$ -spaces (maybe because it's not enough? Maybe like weak homotopy equivalences we require zigzag pattern to define an equivalence relation).

Theorem 3.8. An A_∞ -map $f : X \rightarrow Y$ of $\mathbf{K}_{\mathbb{R}}$ -spaces is a weak group completion, if it's equivalent under a chain of A_∞ -maps to the map $\eta : M \rightarrow \Omega BM$ of some topological monoid M .

Definition 3.9. A Hopf G -space Y is grouplike, if each $\pi_0(Y^H)$ is a group.

For G -equivariant version, let X, Y be homotopy associative / commutative Hopf G -space, with Y being grouplike, and with $f : X \rightarrow Y$ be a Hopf G -map. Then, $f^H : X^H \rightarrow Y^H$ is a group completion for all subgroups H of G .

If X is a $\mathbf{K}_{\mathbb{R}}$ - G -space, and $H \leq G$ is a subgroup, then X^H inherits a $\mathbf{K}_{\mathbb{R}}$ -action. (Q: Can we think of this as X^H maps back to itself under the Hopf space structure?)

Definition 3.10. A map $f : X \rightarrow Y$ of $\mathbf{K}_{\mathbb{R}} - G$ -spaces is a weak group completion, if f^H is a weak group completion for all H . (For instance, if f is a group completion).

Here, we also consider the adjunction $(\Sigma^V \dashv \Omega^V)$ (where $S^V :=$ the one-point compactification of V into a sphere, endowed with a G -action that fixes ∞ point).

Theorem 3.11. (The Equivariant Approximation Theorem) Let V be a representation of G . If X is G -connected (meaning each subgroup $H \leq G$ has X^H being path-connected), then the map

$$\alpha_V : \mathbf{K}_V X \xrightarrow{\mathbf{K}_V \eta} \mathbf{K}_V \Omega^V \Sigma^V X \xrightarrow{\theta} \Omega^V \Sigma^V X$$

is an equivalence (as G -weak equivalence).

If V contains a trivial representation ($\cong \mathbb{R}$), then it's a weak group completion. If V contains a copy of $\mathbb{R}^2$, then α_V is a group completion.

This I think is a way of realizing how to calculate homotopy / homology using the weak equivalence. Similar definitions can be given for compact lie groups, and when extending V to contain $\mathbb{R}^\infty$ -representation some machineries can be extended to that.

3.3 Recognition Principle for V -fold Loop Spaces

3.4 E_∞ G -Operad

Recall that for nonequivariant Σ -operad, each $\mathcal{C}(j)$ is a contractible Σ_j -space